

If κ is a representation of $\mathbf{J}_\theta$ extending η , any other extension of η to $\mathbf{J}_\theta$ has the form $\kappa\xi$ for a unique character ξ of $\mathbf{J}_\theta$ trivial on $\mathbf{J}^1$. More generally, the map

$$(3.4) \quad \tau \mapsto \kappa \otimes \tau$$

is a bijection between isomorphism classes of irreducible representations of $\mathbf{J}_\theta$ trivial on $\mathbf{J}^1$ and isomorphism classes of irreducible representations of $\mathbf{J}_\theta$ whose restriction to $\mathbf{J}^1$ contains η .

4.

In this section, we go back to the group $G = \mathrm{GL}_n(D)$ of §2.1 and fix a cuspidal representation π of G with non-cuspidal transfer to $\mathrm{GL}_{2n}(F)$. In particular, as explained in Remark 2.1, the integer n is odd. The main result of this section is the following proposition.

Proposition 4.1. — *There are a maximal simple stratum $[\mathfrak{a}, \beta]$ in A and a maximal simple character $\theta \in \mathcal{C}(\mathfrak{a}, \beta)$ contained in π such that*

- (1) *the group $H^1(\mathfrak{a}, \beta)$ is stable by σ and $\theta \circ \sigma = \theta^{-1}$,*
- (2) *the order $\mathfrak{a}$ is stable by $*$ and β is invariant by $*$.*

4.1. — Let Θ denote the endoclass of π . Since π contains any maximal simple character of $\mathcal{C}(G)$ of endoclass Θ , it suffices to prove the existence of a maximal simple stratum $[\mathfrak{a}, \beta]$ in A and a character $\theta \in \mathcal{C}(\mathfrak{a}, \beta)$ of endoclass Θ satisfying Conditions (1) and (2) of Proposition 4.1.

Since the Jacquet–Langlands transfer of π is non-cuspidal, it follows from §2.3 that this transfer is of the form $\mathrm{St}_2(\tau)$ for a cuspidal irreducible representation τ of $\mathrm{GL}_n(F)$. By Dotto [12], the representations π and τ have the same endoclass. It follows that the degree of Θ divides n .

4.2. — Let d denote the degree of Θ . Thanks to §4.1, it is an odd integer dividing n .

Let σ_0 be the involution $x \mapsto {}^{\natural}x^{-1}$ on $\mathrm{GL}_d(F)$ where ${}^{\natural}$ denotes the transpose with respect to the antidiagonal. The fixed points of $\mathrm{GL}_d(F)$ by σ_0 is a split orthogonal group.

By [34] Theorem 4.1, there are a maximal simple stratum $[\mathfrak{a}_0, \beta]$ in $\mathbf{M}_d(F)$ and a maximal simple character $\theta_0 \in \mathcal{C}(\mathfrak{a}_0, \beta)$ of endoclass Θ such that

- the group $H^1(\mathfrak{a}_0, \beta)$ is stable by σ_0 and $\theta_0 \circ \sigma_0 = \theta_0^{-1}$,
- the order $\mathfrak{a}_0$ is stable by ${}^{\natural}$ and β is invariant by ${}^{\natural}$.

Write E for the sub- F -algebra $F[\beta] \subseteq \mathbf{M}_d(F)$. It is made of ${}^{\natural}$ -invariant matrices. Its centralizer in $\mathbf{M}_d(F)$ is equal to E itself. The intersection $\mathfrak{a}_0 \cap E$ is $\mathcal{O}_E$, the ring of integers of E .

4.3. — Let us write $n = md$. We identify $\mathbf{M}_n(F)$ with $\mathbf{M}_m(\mathbf{M}_d(F))$ and E with its diagonal image in $\mathbf{M}_n(F)$. The centralizer of E in $\mathbf{M}_n(F)$ is thus $\mathbf{M}_m(E)$.

Now consider $\mathbf{M}_n(F)$ as a sub- F -algebra of A . The centralizer B of E in A is equal to $\mathbf{M}_m(C)$ where C is an E -algebra isomorphic to $E \otimes_F D$. Since the degree d of E over F is odd, C is a non-split quaternion E -algebra. Denote by $*_B$ the anti-involution on B analogous to (2.2).

Proposition 4.2. — *The restriction of $*$ to B is equal to $*_B$.*